# Harshavardhan Reddy

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## Summary

Full-stack developer and Computer Science student. Ships React and Next.js frontends against Python and Node.js APIs backed by PostgreSQL, owning features end to end from protocol and schema to deployment — including real-time systems over WebSockets and Server-Sent Events, and production sites that are live and maintained.

## Technologies

**Languages:** TypeScript/JavaScript, Python, C/C++, Java, SQL, Bash
**Frontend:** React, Next.js (App Router), Vite, TailwindCSS, TanStack Query, React Hook Form, Framer Motion
**Backend:** FastAPI, Node.js, Fastify, Hono, Flask, REST APIs, WebSockets, Server-Sent Events, JWT and Supabase Auth, Zod
**Databases:** PostgreSQL, TimescaleDB, Supabase, MongoDB, Prisma ORM, Redis, SQLite
**Tools & DevOps:** Docker, Git, GitHub Actions, Playwright, Vercel, Cloudflare, AWS, Linux, Prometheus, Grafana

## Projects

**Meridian — Satellite Ground Station Control Platform** 2026 | site · repo

- Final-year project, software platform near complete: predicts which satellite passes are worth receiving, schedules them across a station network, and reports measured reliability; leading a 3-person team
- Six-module platform (orbit propagation, yield prediction, scheduler, registry, observations, reliability) on FastAPI and PostgreSQL/TimescaleDB, with a dashboard that shows why each pass was chosen or skipped
- Designed and froze MSP 0.1, an open four-endpoint HTTP/JSON protocol any station can implement — small enough for a microcontroller, resilient to dropped connections and reboots

**Alisu — Multilingual Voice Assistant for a Citizen Helpline** Mar 2026 | live · repo

- Full-stack voice agent for Karnataka's 1092 helpline handling calls in Kannada, Hindi, and English — files complaints, routes them to the right department, and returns a reference ID
- Fastify + TypeScript server streaming PCM16 audio over WebSockets, with a React + Vite dashboard rendering live transcripts and complaint cards; Twilio telephony, streaming ASR, and a 5-step conversation state machine

**Bathala Enterprises — Real-Estate Listings & Enquiry Platform** Sep 2025 | live · repo

- Built and maintain a live Next.js 16 / React 19 / TypeScript site for a real-estate brokerage — listings with slug routing and ISR, and a Supabase-auth-gated admin dashboard with CRUD for properties, services, and an enquiry inbox
- Enquiry capture with React Hook Form and Zod, Resend notifications, and a Gemini FAQ chatbot; shipped through GitHub Actions running lint, typecheck, build, and Playwright suites

**NetWatch — Local Network Intelligence Platform** Jul 2025 | repo

- Live packet capture and analysis streamed to a browser SPA over Flask REST + Server-Sent Events, with local-first SQLite storage; led a 5-person team and exhibited at CII Innoverge 2026, Bengaluru

## Education

**Nettur Technical Training Foundation**, Bengaluru — June 2024 – Present
Diploma in Computer Technology and IT Infrastructure — *Expected Graduation: 2027*

- CGPA: 9.38/10 — first rank in class
- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, OOP, Software Testing

## Achievements & Certifications

- **1st Prize — Flux of Nature hackathon**, Jain University, Bengaluru (Apr 2026); owned the software layer of an autonomous agriculture system — sensor ingestion, threshold logic, actuator control
- **Round 2 of 3 — Government of Karnataka hackathon**, HackerEarth (2026), for Alisu
- **AWS Cloud Operations** (Jul 2026, verify) · **CCNA: Enterprise Networking & Security** (Sep 2025)